

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - IX

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Class :SY BSC IT	Batch : 01
Date of Assignment :30/9/2026	Date/Time of Submission :30/9/2026

1. Import a CSV dataset using Pandas.

```
import pandas as pd
```

```
df = pd.read_csv("movie_ratings.csv")

print("Original Dataset:")
display(df)
```

Original Dataset:

	Movie	Genre	Release_Year	Rating	Votes	Runtime	Revenue	Language
0	3 Idiots	Drama	2009	8.4	410000	170	460	Hindi
1	Dangal	Sports	2016	8.3	210000	161	2024	Hindi
2	PK	Comedy	2014	8.1	190000	153	854	Hindi
3	Taare Zameen Par	Drama	2007	8.3	180000	165	135	Hindi
4	Sholay	Action	1975	8.1	180000	204	350	Hindi
...	...	...	...	...	...	...	...	...
104	Piku	Comedy	2015	7.6	70000	123	1400	Hindi
105	Bareilly Ki Barfi	Comedy	2017	7.5	50000	116	580	Hindi
106	Ludo	Comedy	2020	7.6	50000	150	0	Hindi
107	Lootera	Romance	2013	7.3	50000	136	290	Hindi
108	Sonchiriya	Action	2019	7.9	30000	143	20	Hindi

109 rows × 8 columns

2. Find missing values in the dataset.

```
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:
Movie      0
Genre      0
Release_Year  0
Rating     0
Votes      0
Runtime     0
Revenue     0
Language    0
dtype: int64
```

3. Replace missing numerical values with the mean.

```
numeric_columns = df.select_dtypes(include='number').columns

for column in numeric_columns:
    df[column] = df[column].fillna(df[column].mean())
```

```
print("Missing numerical values replaced with mean.")
```

Missing numerical values replaced with mean.

4. Remove duplicate records.

```
print("Duplicate records before removal:", df.duplicated().sum())

df = df.drop_duplicates()

print("Duplicate records after removal:", df.duplicated().sum())
```

Duplicate records before removal: 0
Duplicate records after removal: 0

5. Display the cleaned dataset.

```
print("Cleaned Dataset:")
display(df)
```

Cleaned Dataset:

	Movie	Genre	Release_Year	Rating	Votes	Runtime	Revenue	Language
0	3 Idiots	Drama	2009	8.4	410000	170	460	Hindi
1	Dangal	Sports	2016	8.3	210000	161	2024	Hindi
2	PK	Comedy	2014	8.1	190000	153	854	Hindi
3	Taare Zameen Par	Drama	2007	8.3	180000	165	135	Hindi
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...	...	...	...	...	...	...	...	...
104	Piku	Comedy	2015	7.6	70000	123	1400	Hindi
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